SKILL-10: Global Exception Handling using @ControllerAdvice

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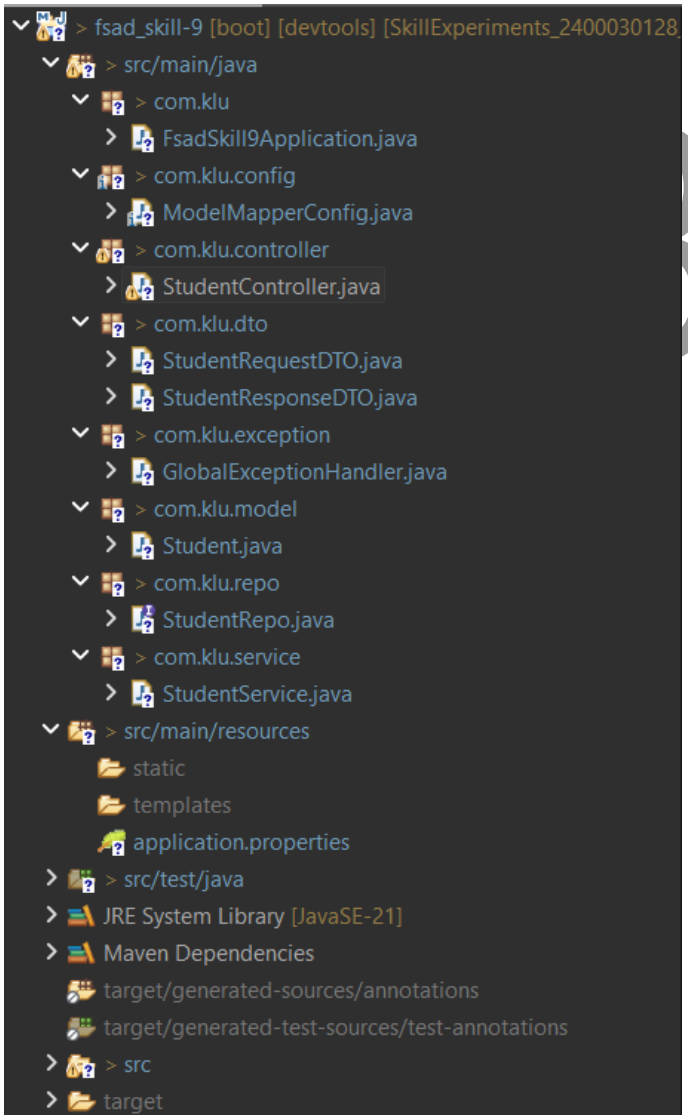
Section: 2

Scenario:

A university application provides student details through REST APIs. When users enter an invalid student ID, the system shows technical error messages that are difficult for users to understand. To improve usability, the application should return clear, readable, and user-friendly error responses through centralized exception handling. Your task is to implement Global Exception Handling using @ControllerAdvice and create custom exceptions for invalid student inputs.

Structure:



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Codes:

Student.java:

```
package com.klu.model;


import jakarta.persistence.Entity;
import jakarta.persistence.GeneratedValue;
import jakarta.persistence.GenerationType;
import jakarta.persistence.Id;
import jakarta.persistence.Table;
import lombok.AllArgsConstructor;
import lombok.Data;
import lombok.NoArgsConstructor;


@Data
@AllArgsConstructor
@NoArgsConstructor
@Entity
@Table(name="student_skill10")
public class Student {
  @Id
  @GeneratedValue(strategy = GenerationType.IDENTITY)
  private Long id;      // lowercase 'id'
  private String name;
  private String email;
  private String branch;
  private String course;
  private Double fees;   // wrapper type for null safety
```

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```
    private String password;
}
```

StudentRepo.java:

```
package com.klu.repo;

import org.springframework.data.jpa.repository.JpaRepository;
import org.springframework.stereotype.Repository;

import com.klu.model.Student;

@Repository
public interface StudentRepo extends JpaRepository<Student, Long>{

}
```

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StudentService.java:

```
package com.klu.service;

import java.util.List;
import java.util.stream.Collectors;

import org.modelmapper.ModelMapper;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Service;

import com.klu.dto.StudentRequestDTO;
import com.klu.dto.StudentResponseDTO;
import com.klu.model.Student;
import com.klu.repo.StudentRepo;

@Service
public class StudentService {
```

```
@Autowired
public StudentRepo studentRepo;
@Autowired
public ModelMapper modelMapper;

//create student
public StudentResponseDTO addStudent(StudentRequestDTO reqdto) {
 //step-1: convert dto to entity
 Student entity = modelMapper.map(reqdto, Student.class);
 //step-2: assign default password in the entity
 entity.setPassword("klu123");

 return modelMapper.map(studentRepo.save(entity),StudentResponseDTO.class);
}

//Read all student details
public List<StudentResponseDTO> getAllStudents() {
        List<Student> list = studentRepo.findAll();
        return list.stream()
                .map(student -> modelMapper.map(student, StudentResponseDTO.class))
                .collect(Collectors.toList());
        }

//Read Specific student
public StudentResponseDTO getStudentById(long id) {
        Student s = studentRepo.findById(id).orElseThrow();
        return modelMapper.map(s, StudentResponseDTO.class);
}

//delete specific student
public String deleteStudent(Long id) {
```

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```
        studentRepo.findById(id).orElseThrow();
        studentRepo.deleteById(id);
        return "Student Deleted Successfully";
    }

    //update Student
    public StudentResponseDTO updateStudent(Long id, StudentRequestDTO s) {
        Student existedStudent = studentRepo.findById(id).orElseThrow();
        existedStudent.setName(s.getName());
        existedStudent.setEmail(s.getEmail());
        existedStudent.setCourse(s.getCourse());
        existedStudent.setBranch(s.getBranch());
        existedStudent.setFees(s.getFees());
        Student updated = studentRepo.save(existedStudent);
        return modelMapper.map(updated, StudentResponseDTO.class);

    }
}
```

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StudentController.java:

```
package com.klu.controller;

import java.util.List;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.http.HttpStatus;
import org.springframework.http.ResponseEntity;
import org.springframework.web.bind.annotation.CrossOrigin;
import org.springframework.web.bind.annotation.DeleteMapping;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.PathVariable;
import org.springframework.web.bind.annotation.PostMapping;
```

```
import org.springframework.web.bind.annotation.RequestBody;
import org.springframework.web.bind.annotation.RequestMapping;
import org.springframework.web.bind.annotation.RestController;
import org.springframework.web.service.annotation.PutExchange;

import com.klu.dto.StudentRequestDTO;
import com.klu.dto.StudentResponseDTO;
import com.klu.service.StudentService;

import jakarta.validation.Valid;
import org.springframework.web.bind.annotation.PutMapping;


@RestController
@RequestMapping("/stud")
@CrossOrigin("*")
public class StudentController {
 @Autowired
 private StudentService studentService;

 @PostMapping("/addStudent")
 public ResponseEntity<StudentResponseDTO> addStudent(@Valid @RequestBody
StudentRequestDTO dto){
  return new ResponseEntity<>(studentService.addStudent(dto),HttpStatus.CREATED);
 }

 @GetMapping("/getAllStudents")
 public ResponseEntity<List<StudentResponseDTO>> getAllStudents(){
        return ResponseEntity.ok(studentService.getAllStudents());
 }
 @GetMapping("/getStudentById/{id}")
```

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```
 public ResponseEntity<StudentResponseDTO> getStudentById(@PathVariable Long id){
        return ResponseEntity.ok(studentService.getStudentById(id));
 }
 @DeleteMapping("/deleteStudent/{id}")
 public ResponseEntity<String> deleteStudent(@PathVariable Long id){
        return ResponseEntity.ok(studentService.deleteStudent(id));
 }

 @PutMapping("/updateStudent/{id}")
 public ResponseEntity<StudentResponseDTO> updateStudent(@PathVariable Long id,
                                @Valid @RequestBody StudentRequestDTO dto) {
   StudentResponseDTO updatedStudent = studentService.updateStudent(id, dto);
   return ResponseEntity.ok(updatedStudent);
 }
}
```

ModelMapperConfig.java:

```
package com.klu.config;

import org.modelmapper.ModelMapper;
import org.springframework.context.annotation.Bean;
import org.springframework.context.annotation.Configuration;

@Configuration
public class ModelMapperConfig {
 @Bean
 public ModelMapper getModelMapper() {
  return new ModelMapper();
 }
}
```

GlobalExceptionHandler.java:

```
package com.klu.exception;
```

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```
import java.util.HashMap;
import java.util.Map;

import org.springframework.web.bind.MethodArgumentNotValidException;
import org.springframework.web.bind.annotation.ExceptionHandler;
import org.springframework.web.bind.annotation.RestControllerAdvice;

@RestControllerAdvice
public class GlobalExceptionHandler {
	@ExceptionHandler(MethodArgumentNotValidException.class)
	public Map<String, String>
handleMethodArgumentNotFoundException(MethodArgumentNotValidException e){
		Map<String,String> errors = new HashMap<>();
		e.getBindingResult().getFieldErrors()
				.forEach(student -> errors.put(student.getField(),
student.getDefaultMessage()));

		return errors;

	}

	 @ExceptionHandler(Exception.class)
	 public String handleInternalServerException(Exception e) {
	  return e.getMessage();
	 }
}
```

StudentResponseDTO.java:

```
package com.klu.dto;

import lombok.AllArgsConstructor;
import lombok.Data;
```

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```
import lombok.NoArgsConstructor;

@Data
@NoArgsConstructor
@AllArgsConstructor
public class StudentResponseDTO {
  private Long id;
  private String name;
  private String email;
  private String branch;
  private String course;
  private double fees;
}
```

StudentRequestDTO.java:

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```
package com.klu.dto;

import jakarta.validation.constraints.Email;
import jakarta.validation.constraints.NotBlank;
import jakarta.validation.constraints.NotEmpty;
import jakarta.validation.constraints.Positive;
import lombok.AllArgsConstructor;
import lombok.Data;
import lombok.NoArgsConstructor;

@Data
@AllArgsConstructor
@NoArgsConstructor
public class StudentRequestDTO {
  @NotEmpty(message = "Student name is required")
  private String name;
  @Email(message = "Email must have proper format")
```

```
  private String email;
  @NotBlank(message = "Branch must not be null")
  private String branch;
  @NotBlank(message = "Course must not be null")
  private String course;
  @Positive(message = "Fees must be positive number")
  private double fees;
}
```

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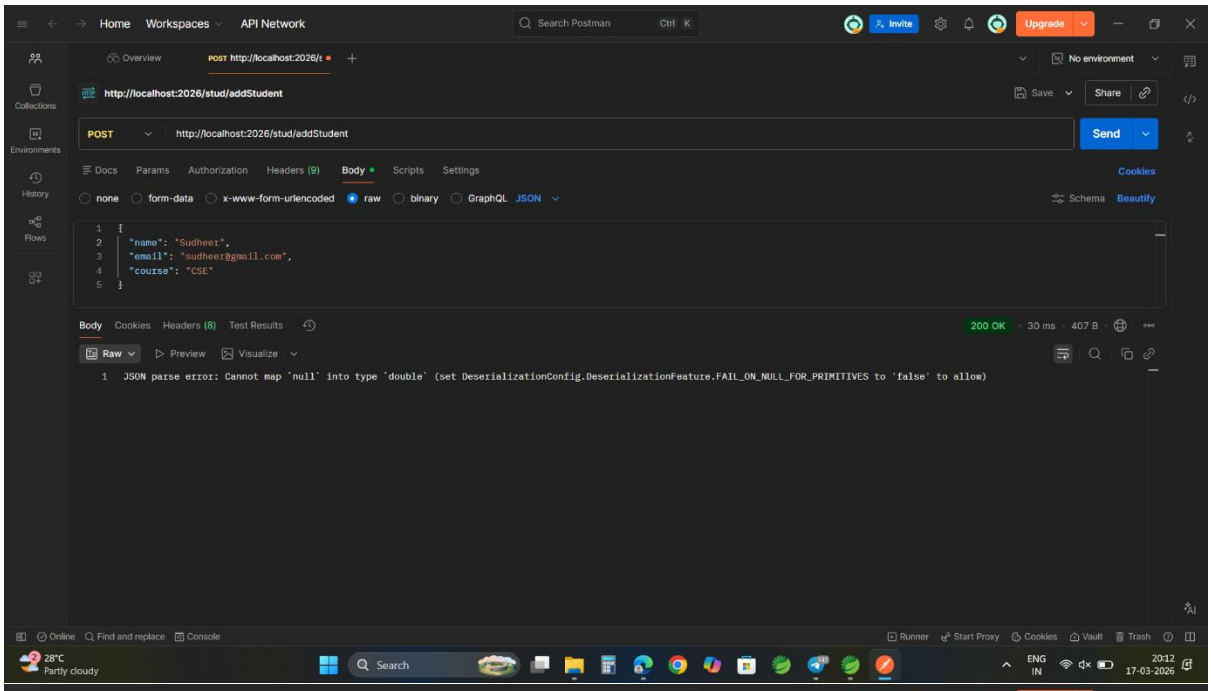
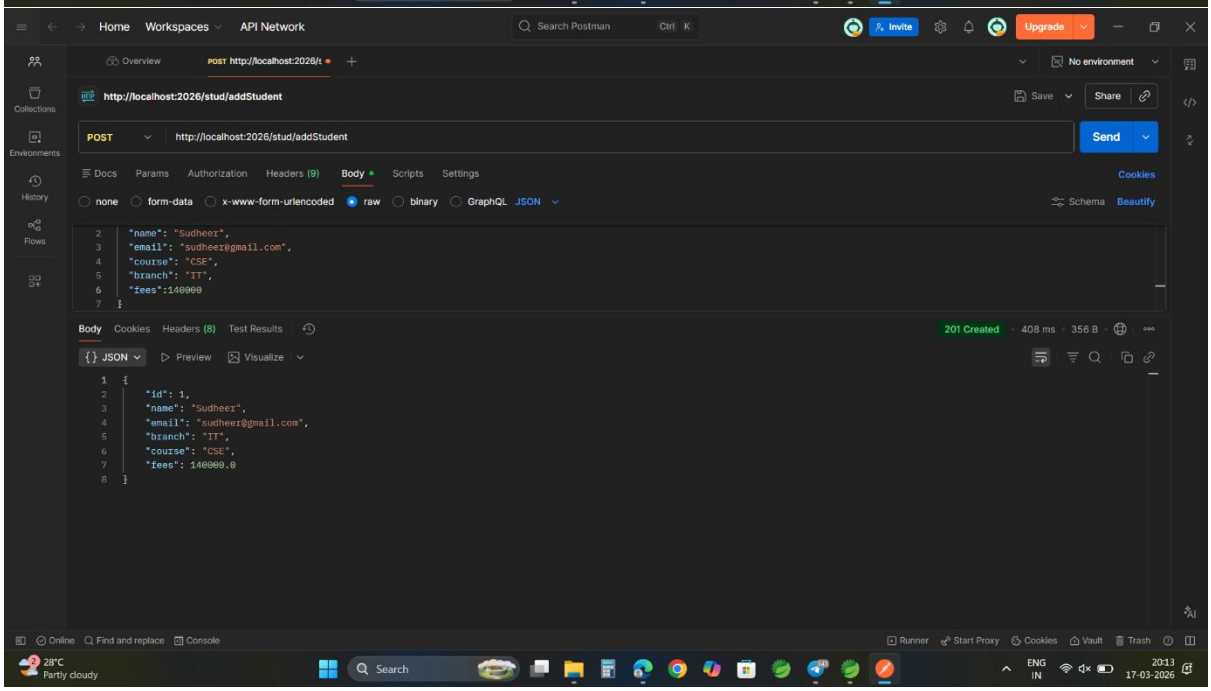
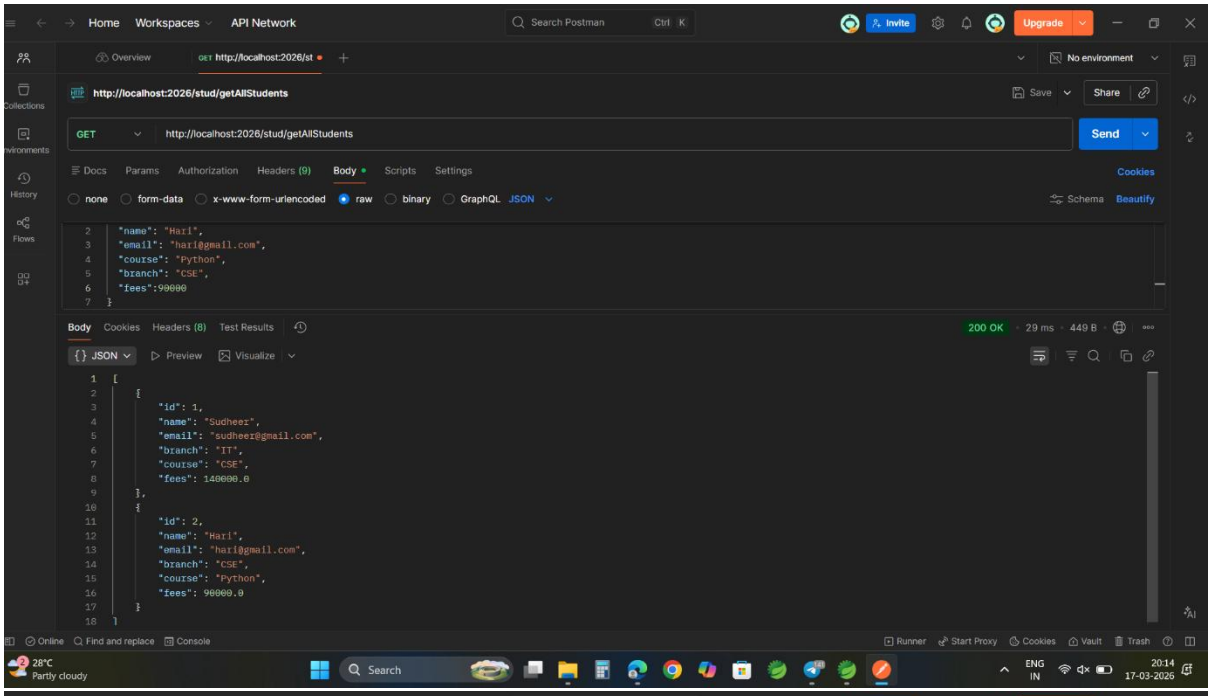
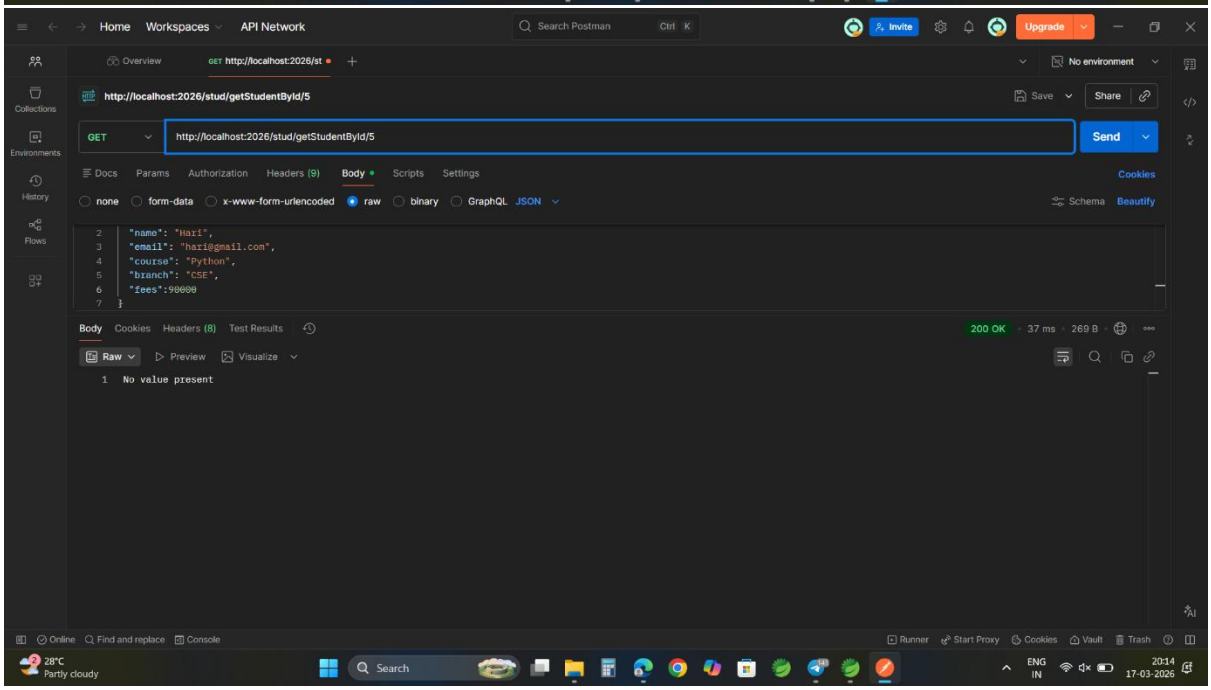
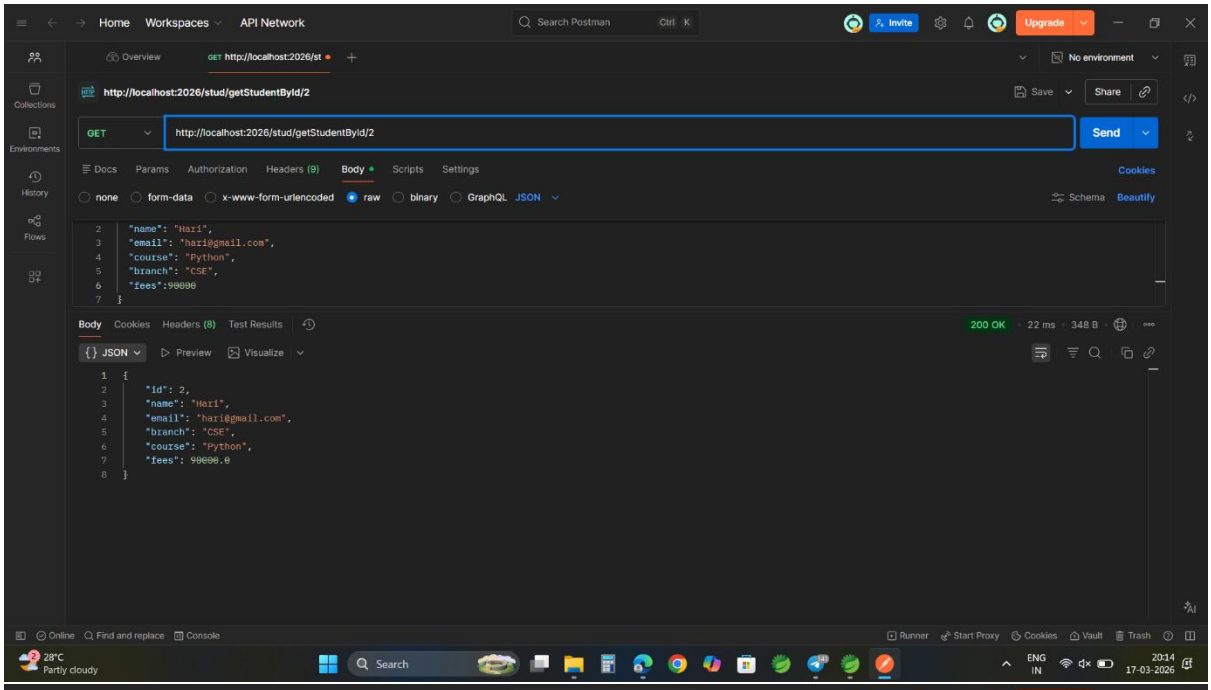
OUTPUTS:

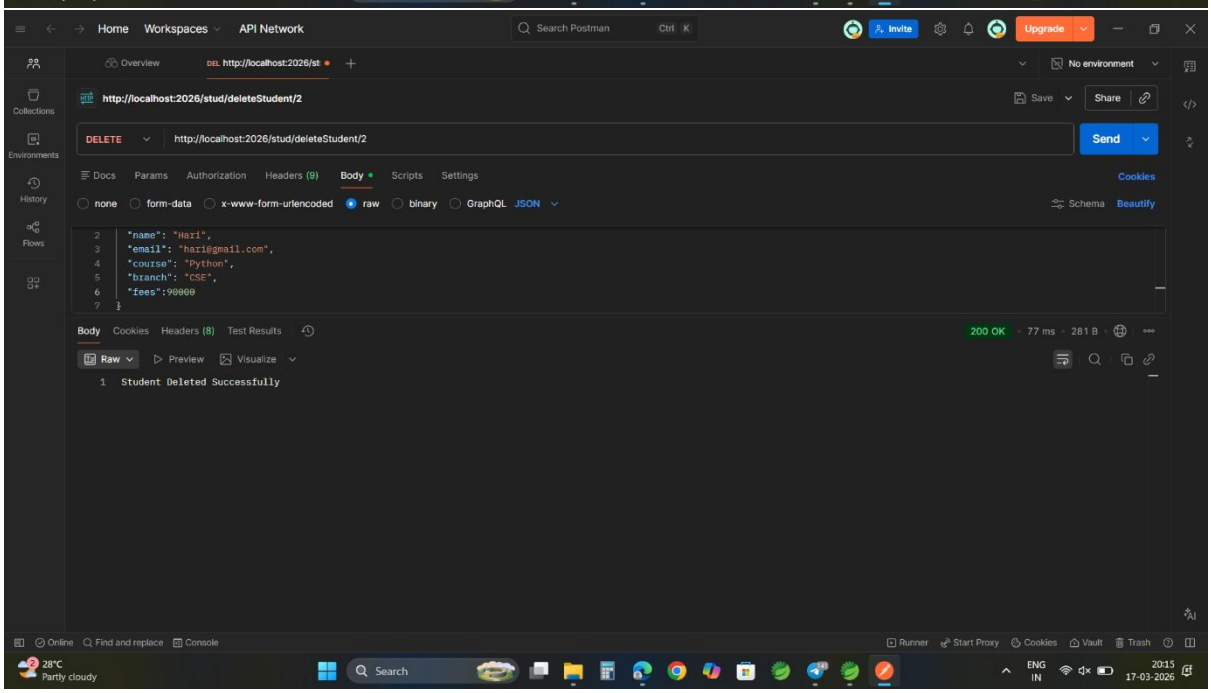

Home Workspaces API Network
Search Postman Ctrl K
Invite
Upgrade
Overview
POST http://localhost:2026/s
No environment
http://localhost:2026/stud/addStudent
Save
Share
POST
http://localhost:2026/stud/addStudent
Send
Docs Params Authorization Headers (9) Body Scripts Settings
Cookies
none form-data x-www-form-urlencoded raw binary GraphQL JSON
Schema Beautify
{
"name": "Sudheer",
"email": "sudheer@gmail.com",
"course": "CSE"
}
Body Cookies Headers (8) Test Results
200 OK 30 ms 407 B
Raw Preview Visualize
JSON parse error: Cannot map `null` into type `double` (set DeserializationConfig.DeserializationFeature.FAIL_ON_NULL_FOR_PRIMITIVES to 'false' to allow)
Online Find and replace Console
Runner Start Proxy Cookies Vault Trash
28°C Partly cloudy
Search
ENG IN
20:12 17-03-2026

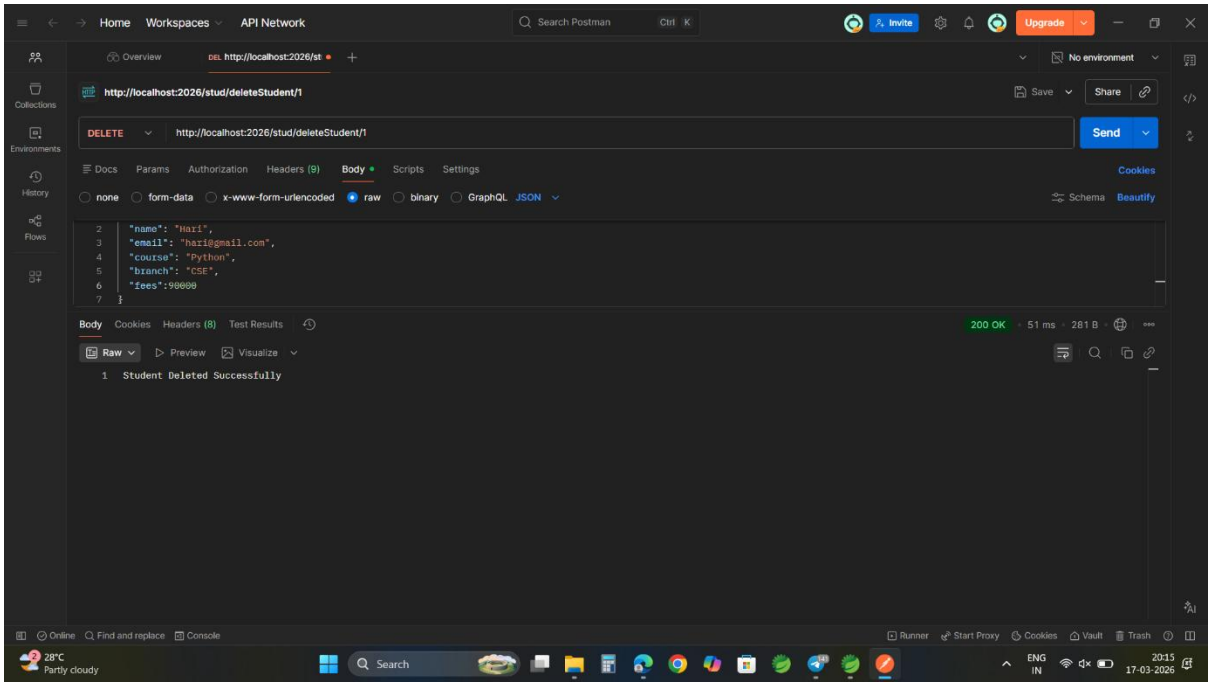

Home Workspaces API Network
Search Postman Ctrl K
Invite
Upgrade
Overview
POST http://localhost:2026/s
No environment
http://localhost:2026/stud/addStudent
Save
Share
POST
http://localhost:2026/stud/addStudent
Send
Docs Params Authorization Headers (9) Body Scripts Settings
Cookies
none form-data x-www-form-urlencoded raw binary GraphQL JSON
Schema Beautify
"name": "Sudheer",
"email": "sudheer@gmail.com",
"course": "CSE",
"branch": "IT",
"fees":140000
}
Body Cookies Headers (8) Test Results
201 Created 408 ms 356 B
JSON Preview Visualize
{
"id": 1,
"name": "Sudheer",
"email": "sudheer@gmail.com",
"branch": "IT",
"course": "CSE",
"fees": 140000.0
}
Online Find and replace Console
Runner Start Proxy Cookies Vault Trash
28°C Partly cloudy
Search
ENG IN
20:13 17-03-2026


Home Workspaces API Network
Search Postman Ctrl K
Invite
Upgrade
Overview
GET http://localhost:2026/st
No environment
http://localhost:2026/stud/getAllStudents
Save
Share
GET
http://localhost:2026/stud/getAllStudents
Send
Docs Params Authorization Headers (9) Body Scripts Settings
Cookies
none form-data x-www-form-urlencoded raw binary GraphQL JSON
Schema Beautify
"name": "Hari",
"email": "hari@gmail.com",
"course": "Python",
"branch": "CSE",
"fees":90000
Body Cookies Headers (8) Test Results
200 OK 29 ms 449 B
JSON Preview Visualize
[
{
"id": 1,
"name": "Sudheer",
"email": "sudheer@gmail.com",
"branch": "IT",
"course": "CSE",
"fees": 140000.0
},
{
"id": 2,
"name": "Hari",
"email": "hari@gmail.com",
"branch": "CSE",
"course": "Python",
"fees": 90000.0
}
]
Online Find and replace Console
Runner Start Proxy Cookies Vault Trash
28°C Partly cloudy
Search
ENG IN
20:14 17-03-2026


Home Workspaces API Network
Search Postman Ctrl K
Invite
Upgrade
Overview
GET http://localhost:2026/st
No environment
http://localhost:2026/stud/getStudentById/5
Save
Share
GET
http://localhost:2026/stud/getStudentById/5
Send
Docs Params Authorization Headers (9) Body Scripts Settings
Cookies
none form-data x-www-form-urlencoded raw binary GraphQL JSON
Schema Beautify
"name": "Hari",
"email": "hari@gmail.com",
"course": "Python",
"branch": "CSE",
"fees":90000
Body Cookies Headers (8) Test Results
200 OK 37 ms 269 B
Raw Preview Visualize
No value present
Online Find and replace Console
Runner Start Proxy Cookies Vault Trash
28°C Partly cloudy
Search
ENG IN
20:14 17-03-2026


Home
Workspaces
API Network
Search Postman
Invite
Upgrade
Overview
GET http://localhost:2026/st
No environment
http://localhost:2026/stud/getStudentById/2
Save
Share
GET
http://localhost:2026/stud/getStudentById/2
Send
Docs
Params
Authorization
Headers (9)
Body
Scripts
Settings
Cookies
none
form-data
x-www-form-urlencoded
raw
binary
GraphQL
JSON
Schema
Beautify
"name": "Hari",
"email": "hari@gmail.com",
"course": "Python",
"branch": "CSE",
"fees":90000
Body
Cookies
Headers (8)
Test Results
200 OK
22 ms
348 B
JSON
Preview
Visualize
"id": 2,
"name": "Hari",
"email": "hari@gmail.com",
"branch": "CSE",
"course": "Python",
"fees": 90000.0
Online
Find and replace
Console
Runner
Start Proxy
Cookies
Vault
Trash
28°C
Partly cloudy
Search
ENG
IN
20:14
17-03-2026


Home
Workspaces
API Network
Search Postman
Invite
Upgrade
Overview
DEL http://localhost:2026/st
No environment
http://localhost:2026/stud/deleteStudent/2
Save
Share
DELETE
http://localhost:2026/stud/deleteStudent/2
Send
Docs
Params
Authorization
Headers (9)
Body
Scripts
Settings
Cookies
none
form-data
x-www-form-urlencoded
raw
binary
GraphQL
JSON
Schema
Beautify
"name": "Hari",
"email": "hari@gmail.com",
"course": "Python",
"branch": "CSE",
"fees":90000
Body
Cookies
Headers (8)
Test Results
200 OK
77 ms
281 B
Raw
Preview
Visualize
Student Deleted Successfully
Online
Find and replace
Console
Runner
Start Proxy
Cookies
Vault
Trash
28°C
Partly cloudy
Search
ENG
IN
20:15
17-03-2026



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GITHUB LINK:

https://github.com/nagasubrahmanyammeka/SkillExperiments_2400030128_FSAD_Sec2/tree/main/SKILL-9